

Vol 0 Chapter 7 — The Operator Zoo

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Chapter 7 — The Operator Zoo

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Every quantum mechanics textbook begins by introducing operators. Position $\hat{X}$ multiplies wavefunctions by the coordinate. Momentum $\hat{P}$ is the derivative. Angular momentum, spin, the Hamiltonian, parity, time reversal, charge conjugation, the number operator, the Bell-CHSH operator — the inventory is finite, well-known, and accepted as part of the framework’s input. Standard quantum mechanics does not derive these operators; it postulates them. Each operator’s identity, its action on the Hilbert space, and its commutation properties with the other operators are part of what a student must learn before doing physics.

In BST the operators are not postulated. They are *derived* — each from a specific feature of the substrate geometry D_{IV}^5 , acting on the substrate’s canonical Hilbert space $H^2(D_{IV}^5)$ (the Bergman Hilbert space we have been using since Chapter 1), with commutation algebra following from the Lie-algebra structure of $SO_0(5, 2)$. The result is what the team calls the operator zoo: an inventory of about a dozen substrate-native operators, organized by the part of the substrate’s symmetry group that produces them, with their commutators and their physical interpretations built in from the geometry.

This chapter is the inventory chapter. After it, we will be ready to do physics.

7.1 The substrate Hilbert space

Every operator in the zoo acts on the same Hilbert space — Lyra’s SP-31-1 result, paper-grade as of May 2026, established that there is a single canonical choice:

$$\mathcal{H}_{\text{sub}} = H^2(D_{IV}^5),$$

the space of holomorphic functions on D_{IV}^5 that are square-integrable with respect to the Bergman volume. This is the Bergman Hilbert space we met in §1.5 of Chapter 1. It is unique up to normalization (the normalization is the $c_{FK} \cdot \pi^{9/2} = 225$ identity of Chapter 2), it is $SO_0(5, 2)$ -invariant (the symmetry group acts on it by unitary transformations), and it admits a reproducing kernel — the Bergman kernel itself — that we have used throughout this volume.

There are two further structures attached to this Hilbert space that we should name, because operators will use them:

- A per-tick discretization. At each Koons tick (Chapter 3), the substrate's continuous Bergman state is represented by a Reed–Solomon codeword in $GF(128)^k$. The continuous $H^2(D_{IV}^5)$ describes the substrate at the long-time-averaged level; the discrete code-space describes it per cycle. The two are related; the relation is what Lyra's T2429 calls the per-tick discretization corollary.
- A K-type decomposition. Under the action of the isotropy subgroup $SO(5) \times SO(2)$, the Hilbert space splits into irreducible representations called K-types, each labeled by a pair of integer weights. The Wallach 1976 paper we cited in Chapter 1 contains the explicit classification. We will not need the details, but we will need to know that the lowest non-trivial K-type carries Casimir eigenvalue $C_2 = 6$ — the BST primary integer we met in Chapter 2.

Both of these structures will be used by specific operators below. The main fact, though, is the simplest: one Hilbert space, $H^2(D_{IV}^5)$, on which every operator in the zoo acts.

7.2 Position and momentum

We begin with the operators of standard one-particle quantum mechanics — position $\hat{X}$ and momentum $\hat{P}$ — because they are the most familiar, and because the substrate's realization of them is structurally interesting.

The point that bears repeating from Chapter 4 is that position and momentum are *coset operators*, not isotropy operators. They live in the complement $\mathfrak{m}$ to the isotropy in $\mathfrak{so}(5, 2)$, not in the isotropy itself. Concretely:

- Position $\hat{X}$ acts on $f \in H^2(D_{IV}^5)$ by multiplication by the coordinate: $(\hat{X}f)(z) = z \cdot f(z)$. The operator carries the BST integer signature of the position-eigenvalue spectrum, which Elie's K71-ratified result of May 2026 showed to align with the perfect numbers cluster — its spectrum's lowest non-trivial trace is 6, then 28, then 496, then 8128, which is the first part of the famous sequence of perfect numbers from antiquity. This identification (Lyra T2419, Elie's substrate-position-operator discovery, K71 ratified) is one of the more striking number-theoretic links the substrate produces. Position is, structurally, *what counts the perfect numbers*.
- Momentum $\hat{P}$ acts by the Wirtinger derivative: $(\hat{P}f)(z) = -i\partial_z f(z)$. It is the

dual operator to position in the coset directions, self-adjoint with respect to the Bergman inner product. (T2422 makes this precise.)

The canonical commutation relation $[\hat{X}, \hat{P}] = i\hbar$ is what physics students memorize. In BST, it is *not* memorized. It is the structural consequence of the Bergman kernel's reproducing property — the integral identity from Chapter 1 — applied to the coset directions in $\mathfrak{m}$. The Heisenberg relation is a theorem on the substrate, derived in two lines from the kernel.

This is the operational pattern that will repeat throughout this chapter and the rest of the book. Standard quantum mechanics postulates a relation; BST derives the relation from the substrate's geometry. The structural origin is always specifiable, often surprising, and always — when the derivation is followed all the way back — anchored in the BST primary integers.

7.3 Angular momentum and spin

Angular momentum $\hat{L}$ comes from the isotropy itself. The ten generators of $SO(5)$, acting on $H^2(D_{IV}^5)$, are the substrate's angular-momentum operators. The familiar three-dimensional angular momentum of laboratory physics is the restriction to the $SO(3) \subset SO(5)$ subgroup that rotates the three physical spatial dimensions — the dimensions controlled by $N_c = 3$, as we discussed in Chapter 4.

The angular-momentum algebra has the standard structure $[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k$, with eigenvalues $j(j+1)\hbar^2$ for integer or half-integer j . Again the algebra is derived, not postulated: $SO(5)$ is a Lie group, and its Lie algebra automatically supplies these commutators.

Spin $\hat{S}$ comes from the K-type structure of substrate representations. It is not an additional ad-hoc internal degree of freedom; it is the intrinsic representation index of the substrate state under the isotropy's $\text{Pin}(2)$ double cover, the additional discrete structure rank-2 forces (Chapter 4). Half-integer spin (fermions) and integer spin (bosons) correspond to the two sectors of the $\text{Pin}(2)$ double cover, and the substrate's commitment cycle (Chapter 3) acts differently on the two sectors — which is, in turn, the structural origin of the spin-statistics theorem. We will derive spin-statistics carefully in Volume 5; the operator zoo's job here is to register that $\hat{S}$ exists, that its commutators with $\hat{L}$ are zero (orbital and spin angular momentum commute), and that the substrate produces it without external postulation.

7.4 The Hamiltonian

The Hamiltonian $\hat{H}$ — the energy operator and the generator of time evolution — comes not from the isotropy but from the full $SO_0(5, 2)$ symmetry. Time translation is a one-parameter subgroup of $SO_0(5, 2)$ that is not contained in the isotropy at the origin, so the Hamiltonian's generators live in the full group rather than in $SO(5) \times SO(2)$.

Elie's K52a Session 29 (May 2026) established the precise form: the substrate Hamiltonian is the Casimir operator of $SO_0(5, 2)$ acting on the appropriate K-type representation. Its lowest non-trivial eigenvalue equals $C_2 = 6$ — the BST primary integer. The Hamiltonian therefore inherits a structural energy unit from the substrate's primary integers.

What this means physically is that the substrate carries a *natural* energy scale — not Planck's constant alone, but $\hbar \times C_2$ at the lowest non-trivial substrate excitation. Particle masses in Volumes 2 and 3 will turn out to be expressible in this natural unit, with the dimensionless multipliers being BST primary combinations. The proton-to-electron mass ratio $6\pi^5$ that we keep mentioning will be one of these; the result drops out of the Hamiltonian's spectrum on the right K-types.

7.5 Charge and chirality

The $SO(2)$ piece of the isotropy supplies two operators, depending on what it acts on.

Electric charge $\hat{Q}$ is the $SO(2)$ weight on substrate scalar states. The eigenvalue spectrum is integer-valued for ordinary substrate states (leptons and gauge bosons take values $0, \pm 1$) and fractional in units of $1/N_c = 1/3$ for color-bearing substrate states (quarks take values $\pm 1/3, \pm 2/3$). The fractional quantization is not a postulate; it falls out of $N_c = 3$ and the substrate's color sub-structure. Lyra's T2470 (May 22, 2026) made this rigorous: charge quantization follows from $SO(2)$ Weyl integrality crossed with the N_c -fold sub-substrate K-types. Casey's W-56 work on Saturday May 24 first identified the substrate-anchoring; the formal substrate-derivation theorem followed Friday afternoon.

Chirality $\hat{\gamma}^5$ is the $SO(2)$ phase on substrate *spinor* states — the same isotropy factor acting differently because the K-type carries half-integer weight rather than integer weight. Eigenvalues are ± 1 , $(\hat{\gamma}^5)^2 = 1$, and the operator anticommutes with the Dirac operator on massless substrate spinors. Lyra T2471 derives this from the $\text{Pin}(2) \mathbb{Z}_2$ grading and the $SO(2)$ half-weight structure on the substrate's spinor bundle. The chirality operator is the same algebraic object that Volume 5 will identify with the spin-statistics grading — chirality and spin-statistics are structurally the same fact about the substrate.

The same $SO(2)$ factor produces two operators because of how it acts: on scalars it multiplies by a phase whose eigenvalue is electric charge; on spinors it multiplies by a half-weight phase whose eigenvalue is chirality. In standard physics, charge and chirality are independent observables; in BST, they are two faces of one substrate factor.

7.6 Parity, time reversal, charge conjugation

The discrete symmetries — parity $\hat{P}$, time reversal $\hat{T}$, and charge conjugation $\hat{C}$ — each correspond to a specific substrate involution.

Parity $\hat{P}$ is the lift of the Möbius involution on D_{IV}^5 that Chapter 4 introduced. Its square is the identity; its action on position, momentum, angular momentum, and spin is exactly what standard physics expects ($\hat{P}\hat{X}\hat{P} = -\hat{X}$, $\hat{P}\hat{P}\hat{P} = -\hat{P}$ where the latter is a momentum operator and the former a parity; $\hat{P}\hat{L}\hat{P} = +\hat{L}$, $\hat{P}\hat{S}\hat{P} = +\hat{S}$). It anticommutes with chirality: $\hat{P}\hat{\gamma}^5\hat{P} = -\hat{\gamma}^5$. Lyra T2472 (May 22, 2026) provides the formal substrate-derivation. The Möbius-locality argument of Chapter 4 explains why parity is conserved in the strong and electromagnetic sectors but violated in the weak sector.

Time reversal $\hat{T}$ is an anti-unitary operator that reverses the substrate’s commitment-cycle direction. It is the Klein anti-unitary involution applied to the substrate’s cycle structure (Lyra T2433). The anti-unitarity (complex conjugation combined with cycle-direction flip) is what makes time reversal a different kind of operator from parity — it does not commute with most observables in the standard way, and Wigner’s theorem on anti-unitary operators is invoked when we work with it in Volume 5.

Charge conjugation $\hat{C}$ is the $SO(2)$ -weight-negating involution: it sends charge to $-$ charge, particle to antiparticle. Lyra T2434 provides the substrate-derivation theorem.

The composite $\hat{C}\hat{P}\hat{T}$ commutes with every substrate Hamiltonian, in every sector, for structural reasons rooted in $SO_0(5, 2)$ ’s conformal action. This is the famous CPT theorem of standard quantum field theory (Lüders 1954, Pauli 1955), recovered in BST as a structural consequence rather than a derived theorem of axiomatic QFT.

7.7 The substrate-cycle operators

Two further operators come from the substrate’s commitment-cycle structure, rather than from $SO_0(5, 2)$ symmetry directly.

The Bell–CHSH operator $\hat{B}$ acts on bipartite substrate states (the tensor product of two single-system Hilbert spaces) and measures the correlation between substrate commitments at separated locations. Its expectation-value-squared on substrate states is bounded by $\text{Tr}(\hat{B}^2) = 126/16 = 7.875$ — a fact that is essential because the Tsirelson bound of standard quantum mechanics is 8. The difference $8 - 126/16 = 1/8 = 1/2^{N_c}$ is the BST sub-Tsirelson signature: substrate-derived quantum mechanics predicts Bell correlations slightly below the Tsirelson bound, with the gap controlled by $N_c = 3$. This is one of the framework’s sharpest experimental falsifiers — a precision Bell experiment with sufficient statistics could measure the gap directly, and either confirm BST or reject it.

The number operator $\hat{N}$ counts substrate cycles. Its eigenvalues are non-negative integers, and its physical interpretation depends on context — in particle-physics applications it counts particles (a particle being a substrate-cycle structure), in cosmological applications it counts integrated commitments at large scale. The T1922 identification “particles are substrate cycles” sits behind this operator.

7.8 The zoo as inventory

Here is the zoo collected in one table, organized by substrate origin and tagged with the substrate-derivation theorem that places the operator on the inventory.

Operator	Symbol	Substrate origin	Theorem	Notes
Position	$\hat{X}$	Coset direction $\mathfrak{m}$	T2419	Spectrum is the perfect-numbers cluster (K71)
Momentum	$\hat{P}$	Coset direction (dual)	T2422	Wirtinger derivative; $[\hat{X}, \hat{P}] = i\hbar$ from Bergman kernel
Angular momentum	$\hat{L}$	$SO(5)$ factor of isotropy	T2425	Standard $SO(3)$ restriction in laboratory
Spin	$\hat{S}$	$Pin(2)$ double cover of isotropy	T2421	Half-integer and integer sectors
Bell-CHSH	$\hat{B}$	Substrate-cycle bipartite correlator	K66	$\text{Tr}(\hat{B}^2) = 126/16$; sub-Tsirelson by $1/8 = 1/2^{N_c}$
Hamiltonian	$\hat{H}$	$SO_0(5, 2)$ Casimir on K-types	K52a S29	Lowest non-trivial eigenvalue is $C_2 = 6$
Charge	$\hat{Q}$	$SO(2)$ weight (scalar K-type)	T2470	Integers + fractional $\pm 1/3, \pm 2/3$ for color states
Chirality	$\hat{\gamma}^5$	$SO(2)$ weight (spinor K-type)	T2471	± 1 eigenvalues; involution
Parity	$\hat{P}$	Möbius involution lift	T2472	Möbius locality $\rightarrow$ weak-sector violation

Operator	Symbol	Substrate origin	Theorem	Notes
Time reversal	$\hat{T}$	Cycle-reversal anti-unitary	T2433	Klein operator + cycle-direction flip
Charge conjugation	$\hat{C}$	$SO(2)$ weight negation	T2434	Unitary involution
Number	$\hat{N}$	Substrate-cycle counter	T1922 (candidate)	Non-negative integer spectrum

The composite $\hat{C}\hat{P}\hat{T}$ is not a separate operator on the inventory; it is a derived composite whose substrate-level identification gives the CPT theorem.

This is the zoo, in one table.

7.9 What the zoo buys

Three things, in particular.

First, the operator zoo is *finite*. There are about twelve fundamental operators in standard quantum mechanics, and the substrate produces exactly this many — no missing operators (every standard QM observable corresponds to a substrate-native object) and, more strikingly, no extra operators (the substrate does not predict observables that physics has not seen). This was Casey’s #6 standing structural principle, the Substrate Cognition Network Hypothesis’s quieter counterpart — completeness of the operator inventory is a substrate signature.

Second, the commutation algebra is *derived*. Standard quantum mechanics treats $[\hat{X}, \hat{P}] = i\hbar$, $[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k$, $\hat{P}^2 = 1$, and so on as separate inputs. BST produces them from the substrate’s Lie-algebra structure plus the Bergman-kernel reproducing property. The whole commutator structure of quantum mechanics is, in the substrate framework, *one theorem*.

Third, the operators carry the integer-web signature of Chapter 6. Position’s spectrum runs through the perfect numbers. Charge quantizes in units of $1/N_c$. Bell-CHSH’s bound is $\text{Tr}(\hat{B}^2) = 126/16 = (2^8 - \text{rank})/2^4$. The Hamiltonian’s lowest eigenvalue is C_2 . The operators do not sit *next to* the integer web; they sit *inside* it.

7.10 What comes next

Chapter 8 applies Noether’s theorem to the substrate symmetries we have now collected. Each continuous symmetry of $SO_0(5, 2)$ that leaves the Hamiltonian invariant produces a conservation law — energy, momentum, angular momentum, charge, lepton number, baryon number, the lot. The conservation laws are derived

rather than postulated, with the same operational pattern we have used throughout this volume.

Chapter 9 returns to Strong-Uniqueness and proves that no alternative geometry produces the operator zoo we have just inventoried. The zoo is one of the criteria the theorem uses.

Chapter 10 closes the volume with the methodology summary.

Where to look this up: The substrate Hilbert space derivation is Lyra SP-31-1, with the K-type decomposition referring to Wallach's 1976 paper in the *Transactions of the AMS*. Per-operator substrate-derivation theorems are catalogued by T-number in the table above. The Bell-CHSH sub-Tsirelson signature with substrate-mechanism is K66 (audit) and Calibration #17 (trace-level versus max-eigenvalue distinction). The Hamiltonian as $SO_0(5, 2)$ Casimir is Elie K52a Session 29. The perfect-numbers identification of the position spectrum is K71 ratified, with Elie's verification toys 3142 and 3160. The CPT-cluster substrate-derivation is the K85-K86-K87 trio. The Operator Zoo Promotion Ledger is filed at `notes/Operator_Zoo_Promotion_Ledger_v0_1.md` and tracks per-operator tier status; readers wanting the audit-history detail should look there.